

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

END TERM EXAMINATION
SECOND SEMESTER [B.TECH] APRIL - MAY 2019

Paper Code: ETEN-114**Subject: Environmental Studies (Batch 2013 Onwards)****Time: 3 Hours****Maximum Marks: 75**

Note: Attempt all questions as directed. Internal choice is indicated.

Q1 Attempt all the parts of the following questions: **(2.5 × 10 = 25)**

- a) What do you understand by 'sustainable development'? Give one main cause of environmental crisis.
- b) What are the main components of an ecosystem?
- c) What steps can be taken to save biodiversity in the area?
- d) What is 'waste' product in chemical reaction? Give at least two examples.
- e) How toxic elements affect the working of the enzymes?
- f) What are the main air pollutants? What do you understand by the term 'sink' with respect to pollution?
- g) "Dose makes the poison". Comment.
- h) What are photodegradable polymers? Give examples.
- i) What are the three forms of ocean energy?
- j) What do you understand by 'carbon credits'? What are its merits and demerits?

Q2 **(3)**
 a) How does an Ecosystem get established? What do you understand by the term Ecological succession?
(3)
 b) What is the "edge effect" and what is its importance?
(4)
 c) Discuss the consumptive and productive values of biodiversity.
(2.5)
 d) What is the need for public awareness to understand the different environmental issues?

OR

Q3 **(4)**
 a) Explain the ecological pyramid. Is it different from energy pyramid?
(3)
 b) How did the 'Green Revolution' affect the diversity of food plants?
(3)
 c) What do you understand by "ecological niche" of a species?
(2.5)
 d) Why are many environmental problems so difficult to solve? Give at least one example.

Q4 **(4)**
 a) What are green fuels? Why are these considered non polluting and clean fuels?
(4)
 b) How are the minerals classified?
(1.5 × 3 = 4.5)
 c) Write short notes on:
 (i) Limitations of Geothermal energy (ii) Water logging
 (iii) Ocean thermal energy conversion

OR

Q5 **(4)**
 a) What are the different types of energies that are derived from the ocean?
(1.5 × 4 = 6)
 b) Write short notes on the following:
 (i) Land subsidence (ii) Surface and subsurface mining
 (iii) Man induced landslides (iv) Salt water intrusion
(2.5)
 c) What is the consequence of overexploitation of water resources?

P.T.O.

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

- Q6**
- a) How do the green starting materials help to improve the environmental conditions? (2)
 - b) Justify "Prevention is better than cure" in context to green chemistry. (2)
 - c) Illustrate with examples how biopolymers are extracted from living organisms. (2)
 - d) What type of polymers are known as hydro-biodegradable polymers? (1.5)
 - e) What is the chemistry of photochemical smog? (2)
 - f) What do you understand by (i) NOAEL (ii) LOAEL (iii) Bioremediation? (1 × 3 = 3)

OR

- Q7**
- a) Write short notes on: (1.5 × 3 = 4.5)
 - (i) BOD₅ (ii) Activated sludge process (iii) Chemical Smog
 - b) How are hazardous wastes classified? (2)
 - c) What are the biochemical effects of Arsenic? (3)
 - d) What do you understand by 'Atom Economy'? (1.5)
 - e) Discuss the methods used to produce bioplastics. (1.5)
- Q8**
- a) Identify three major mitigation methods to reduce earthquake risk. (3)
 - b) What are the various objectives of EIA (Environment Impact Assessment)? (3)
 - c) Why are ISO:14000 standards revised on 5-year cycle? (3)
 - d) What are the factors responsible for depletion of ozone layer? What are the chemical reactions involved? (3.5)

OR

- Q9**
- a) Distinguish between 'magnitude' and 'intensity' of an earthquake. How are they measured? (4)
 - b) What are the effects of consumerism on present society? (4)
 - c) Write short notes on: (1.5 × 3 = 4.5)
 - (i) Ozone depletion (ii) Climate change (iii) Green Bench
